

Interpolating intuitionistic modal logic, in algebra and duality

Rodrigo Nicolau Almeida – ILLC-University of Amsterdam
reporting on j.w.w. Guran Bezhanishvili, Nick Bezhanishvili, Simon Lemal
Thursday 30 July, 2026

Definability theory

My main interest lies in **definability theory**: what kind of higher-order constructs can be defined in (the algebras of) non-classical logics. **Intuitionistic modal logics** provide a number of good test cases for developing this theory.

My main interest lies in **definability theory**: what kind of higher-order constructs can be defined in (the algebras of) non-classical logics. **Intuitionistic modal logics** provide a number of good test cases for developing this theory.

Properties I am interested in:

1. **Interpolation**: in both deductive form and "Craig" form – intimately related to **amalgamation**.

My main interest lies in **definability theory**: what kind of higher-order constructs can be defined in (the algebras of) non-classical logics. **Intuitionistic modal logics** provide a number of good test cases for developing this theory.

Properties I am interested in:

1. **Interpolation**: in both deductive form and "Craig" form – intimately related to **amalgamation**.
2. **Uniform definability of adjoints**: related to **model completions** and **bisimulation quantifiers**.

My main interest lies in **definability theory**: what kind of higher-order constructs can be defined in (the algebras of) non-classical logics. **Intuitionistic modal logics** provide a number of good test cases for developing this theory.

Properties I am interested in:

1. **Interpolation**: in both deductive form and "Craig" form – intimately related to **amalgamation**.
2. **Uniform definability of adjoints**: related to **model completions** and **bisimulation quantifiers**.
3. **Periodicity of endomorphisms**: Ruitenburg's property, related to **fixed-point elimination**.

My main interest lies in **definability theory**: what kind of higher-order constructs can be defined in (the algebras of) non-classical logics. **Intuitionistic modal logics** provide a number of good test cases for developing this theory.

Properties I am interested in:

1. **Interpolation**: in both deductive form and "Craig" form – intimately related to **amalgamation**.
2. **Uniform definability of adjoints**: related to **model completions** and **bisimulation quantifiers**.
3. **Periodicity of endomorphisms**: Ruitenburg's property, related to **fixed-point elimination**.

Today the focus will be mostly on interpolation, which I will talk about in detail; but I am interested in discussing other properties, and I will survey at the end briefly some of the results we obtained.

Intuitionistic modal logic

Which intuitionistic modal logic?

Languages:

$$\mathcal{L}_{\Box} := p \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \rightarrow \psi \mid \Box\phi.$$

and

$$\mathcal{L}_{\Box, \Diamond} := \mathcal{L}_{\Box} \text{ and } \mid \Diamond\phi.$$

Which intuitionistic modal logic?

Languages:

$$\mathcal{L}_{\Box} := p \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \rightarrow \psi \mid \Box\phi.$$

and

$$\mathcal{L}_{\Box, \Diamond} := \mathcal{L}_{\Box} \text{ and } \mid \Diamond\phi.$$

We will take some basic axioms over IPC, whenever the connectives are in the language:

1. $(N_{\Box}) \Box(p \wedge q) \leftrightarrow \Box p \wedge \Box q$ and $\Box \top$.
2. $(N_{\Diamond}) \Diamond(p \vee q) \leftrightarrow (\Diamond p \vee \Diamond q)$ and $\neg \Diamond \perp$.

Which intuitionistic modal logic?

Languages:

$$\mathcal{L}_{\Box} := p \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \rightarrow \psi \mid \Box\phi.$$

and

$$\mathcal{L}_{\Box, \Diamond} := \mathcal{L}_{\Box} \text{ and } \mid \Diamond\phi.$$

We will take some basic axioms over IPC, whenever the connectives are in the language:

1. $(N_{\Box}) \Box(p \wedge q) \leftrightarrow \Box p \wedge \Box q$ and $\Box \top$.
2. $(N_{\Diamond}) \Diamond(p \vee q) \leftrightarrow (\Diamond p \vee \Diamond q)$ and $\neg \Diamond \perp$.
3. $(C_1) (\Diamond(p \rightarrow q) \wedge \Box p) \rightarrow \Diamond q$;
4. $(C_2) (\Diamond p \rightarrow \Box q) \rightarrow \Box(p \rightarrow q)$.

Which intuitionistic modal logic?

The basic examples are, over $\mathcal{L}_\Box$,

$$\mathbf{iK} := \mathbf{IPC} \oplus \mathbf{N}_\Box$$

Which intuitionistic modal logic?

The basic examples are, over $\mathcal{L}_{\Box}$,

$$iK := IPC \oplus N_{\Box}$$

and over $\mathcal{L}_{\Box, \Diamond}$,

$$IK := iK \oplus N_{\Diamond} \oplus C_1 \oplus C_2,$$

also called *Fischer-Servi logic*.

Which intuitionistic modal logic?

The basic examples are, over $\mathcal{L}_{\Box}$,

$$\mathbf{iK} := \mathbf{IPC} \oplus \mathbf{N}_{\Box}$$

and over $\mathcal{L}_{\Box, \Diamond}$,

$$\mathbf{IK} := \mathbf{iK} \oplus \mathbf{N}_{\Diamond} \oplus \mathbf{C}_1 \oplus \mathbf{C}_2,$$

also called *Fischer-Servi logic*.

CONVENTION (Giessen 2022): for a classical modal logic L ,

$\mathbf{iL}$ means the intuitionistic L -logic in $\mathcal{L}_{\Box}$, with just Normality,

$\mathbf{IL}$ means the bimodal intuitionistic L logic in $\mathcal{L}_{\Box, \Diamond}$, with all axioms.

Which intuitionistic modal logic?

These logics form different families (some examples)

Provability/Topological	Lax operators	Bimodal/First-order
KM	PLL	IK
mHC	DPLL	IS4
iGL	DNL	IS5/MIPC

Which intuitionistic modal logic?

These logics form different families (some examples)

Provability/Topological	Lax operators	Bimodal/First-order
KM	PLL	IK
mHC	DPLL	IS4
iGL	DNL	IS5/MIPC

Axiomatizations later. Key observation: KM, mHC, DNL and all lax logic extensions, contain the "Strength axiom":

$$p \rightarrow \Box p,$$

which is classically trivial.

Which intuitionistic modal logic?

Intuition:

Provability/Topological	Lax operators	Bimodal/First-order
KM	PLL	IK
mHC	DPLL	IS4
iGL	DNL	IS5/MIPC

Which intuitionistic modal logic?

Intuition:

Provability/Topological	Lax operators	Bimodal/First-order
KM	PLL	IK
mHC	DPLL	IS4
iGL	DNL	IS5/MIPC

Intuitionistic modalities – carry modal information that concerns intuitionistic truth.

Modalities seen intuitionistically – What can be said intuitionistically about classical modalities (epistemic, deontic, metaphysical, etc).

Which intuitionistic modal logic?

Intuition:

Provability/Topological	Lax operators	Bimodal/First-order
KM	PLL	IK
mHC	DPLL	IS4
iGL	DNL	IS5/MIPC

Intuitionistic modalities – carry modal information that concerns intuitionistic truth.

Modalities seen intuitionistically – What can be said intuitionistically about classical modalities (epistemic, deontic, metaphysical, etc).

Both have a substantial history:

1. **Intuitionistic modalities**: (Kuznetsov 1985; Kuznetsov and Muravitsky 1986), (Goldblatt 1981), (Visser 1981), (Fairtlough and Mendler 1997), (Esakia 2006);
2. **Modalities intuitionistically**: (Fischer-Servi 1984), (Ewald 1986), (Ono 1977), (Simpson 1994).

Why intuitionistic modal logics?

Many interesting modal phenomena only become visible intuitionistically. And many these modalities are **naturally occurring**:

1. **(Proof theory and provability logic)**: via provability interpretations, the logic KM (Kuznetsov and Muravitsky 1986) and related systems interpret provability in some systems of Heyting arithmetic;
2. **(Locale theory)**: lax operators are nuclei, which capture sublocales in **point-free topology**.
3. **(Topology)**: the (co)derivative modality is captured by the axioms of mHC;
4. **(First-order translation)**: the standard translation translates models of IK to first-order intuitionistic logics.

General recipe and interpolation

General recipe for definability problems

We follow the following general recipe, with adaptations where necessary:

1. Find algebraic semantics for the logic;

General recipe for definability problems

We follow the following general recipe, with adaptations where necessary:

1. Find algebraic semantics for the logic;
2. Translate the definability problem into an algebraic problem.

General recipe for definability problems

We follow the following general recipe, with adaptations where necessary:

1. Find algebraic semantics for the logic;
2. Translate the definability problem into an algebraic problem.
3. Find dual representations of those algebras, which allow for more concrete manipulations;

General recipe for definability problems

We follow the following general recipe, with adaptations where necessary:

1. Find algebraic semantics for the logic;
2. Translate the definability problem into an algebraic problem.
3. Find dual representations of those algebras, which allow for more concrete manipulations;
4. Translate the algebraic problem into a dual problem.

General recipe for definability problems

We follow the following general recipe, with adaptations where necessary:

1. Find algebraic semantics for the logic;
2. Translate the definability problem into an algebraic problem.
3. Find dual representations of those algebras, which allow for more concrete manipulations;
4. Translate the algebraic problem into a dual problem.
5. Solve the dual problem!

General recipe for definability problems

We follow the following general recipe, with adaptations where necessary:

1. Find algebraic semantics for the logic;
2. Translate the definability problem into an algebraic problem.
3. Find dual representations of those algebras, which allow for more concrete manipulations;
4. Translate the algebraic problem into a dual problem.
5. Solve the dual problem!

We will warm up with **Propositional Lax Logic**:

$$\text{PLL} := \text{iK} \oplus (p \rightarrow \Box p) \oplus (\Box \Box p \rightarrow \Box p).$$

For intuitionistic modal logics we work with **Heyting algebras with operators**.

Definition

An algebra $(H, \wedge, \vee, \rightarrow, \Box, 0, 1)$ is called a **modal Heyting algebra** if:

1. $(H, \wedge, \vee, 0, 1)$ is a (distributive) lattice.
2. The following law holds for all $a, b, c \in H$:

$$a \wedge c \leq b \iff c \leq a \rightarrow b.$$

3. $\Box$ is a normal operator: $\Box(a \wedge b) = \Box a \wedge \Box b$ and $\Box 1 = 1$.

There is the usual **algebraic completeness**:

Varieties of Algebras = Extensions of Logics.

For intuitionistic modal logics we work with **Heyting algebras with operators**.

Definition

An algebra $(H, \wedge, \vee, \rightarrow, \Box, 0, 1)$ is called a **modal Heyting algebra** if:

1. $(H, \wedge, \vee, 0, 1)$ is a (distributive) lattice.
2. The following law holds for all $a, b, c \in H$:

$$a \wedge c \leq b \iff c \leq a \rightarrow b.$$

3. $\Box$ is a normal operator: $\Box(a \wedge b) = \Box a \wedge \Box b$ and $\Box 1 = 1$.

There is the usual **algebraic completeness**:

Varieties of Algebras = Extensions of Logics.

Definition (Nuclear Heyting algebra)

A **nuclear Heyting algebra** is a modal Heyting algebra which satisfies:

1. $a \leq \Box a$;
2. $\Box \Box a \leq \Box a$.

For intuitionistic modal logics we work with **Heyting algebras with operators**.

Definition

An algebra $(H, \wedge, \vee, \rightarrow, \Box, 0, 1)$ is called a **modal Heyting algebra** if:

1. $(H, \wedge, \vee, 0, 1)$ is a (distributive) lattice.
2. The following law holds for all $a, b, c \in H$:

$$a \wedge c \leq b \iff c \leq a \rightarrow b.$$

3. $\Box$ is a normal operator: $\Box(a \wedge b) = \Box a \wedge \Box b$ and $\Box 1 = 1$.

There is the usual **algebraic completeness**:

Varieties of Algebras = Extensions of Logics.

Definition (Nuclear Heyting algebra)

A **nuclear Heyting algebra** is a modal Heyting algebra which satisfies:

1. $a \leq j(a)$;
2. $jja \leq ja$.

Two main properties:

Definition (Deductive Interpolation (DIP))

Whenever $\phi(\bar{p}, \bar{q}) \vdash_L \psi(\bar{q}, \bar{r})$ there is some $\chi(\bar{q})$ such that $\phi(\bar{p}, \bar{q}) \vdash_L \chi(\bar{q})$ and $\chi(\bar{q}) \vdash_L \psi(\bar{q}, \bar{r})$.

Two main properties:

Definition (Deductive Interpolation (DIP))

Whenever $\phi(\bar{p}, \bar{q}) \vdash_L \psi(\bar{q}, \bar{r})$ there is some $\chi(\bar{q})$ such that $\phi(\bar{p}, \bar{q}) \vdash_L \chi(\bar{q})$ and $\chi(\bar{q}) \vdash_L \psi(\bar{q}, \bar{r})$.

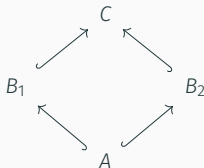
Definition (Craig Interpolation (CIP))

Whenever $\phi(\bar{p}, \bar{q}) \rightarrow \psi(\bar{q}, \bar{r}) \in L$ there is some $\chi(\bar{q})$ such that $\phi(\bar{p}, \bar{q}) \rightarrow \chi(\bar{q}) \in L$ and $\chi(\bar{q}) \rightarrow \psi(\bar{q}, \bar{r}) \in L$.

For these logics, interpolation is very closely related to **amalgamation**.

Definition (Amalgamation)

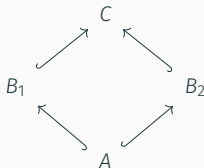
A triple $B_2 \leftarrow A \hookrightarrow B_1$ is called a **V-formation**. An algebra C is called an **amalgam** if the diagram commutes:



For these logics, interpolation is very closely related to **amalgamation**.

Definition (Amalgamation)

A triple $B_2 \leftarrow A \hookrightarrow B_1$ is called a **V-formation**. An algebra C is called an **amalgam** if the diagram commutes:

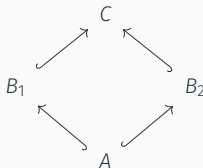


It is a **superamalgam** if whenever $b_1 \leq_C b_2$, there is some $a \in A$ such that $b_1 \leq a$ and $a \leq b_2$.

For these logics, interpolation is very closely related to **amalgamation**.

Definition (Amalgamation)

A triple $B_2 \leftarrow A \hookrightarrow B_1$ is called a **V-formation**. An algebra C is called an **amalgam** if the diagram commutes:



It is a **superamalgam** if whenever $b_1 \leq_C b_2$, there is some $a \in A$ such that $b_1 \leq a$ and $a \leq b_2$.

A class $\mathbf{V}$ has **(super)amalgamation** if each V-formation (super)amalgamates.

In general for our purposes we have the following **bridge theorem**:

Theorem (Maksimova 1979)

A logic L has:

1. *Deductive interpolation if and only if $\text{Alg}(L)$ has amalgamation;*
2. *Craig interpolation if and only if $\text{Alg}(L)$ has superamalgamation.*

CAVEAT: The two properties are often not equivalent. For intuitionistic logic, lax logic, and extensions of mHC, however, there is a **true deduction theorem**:

$$\phi \vdash_L \psi \iff \vdash_L \phi \rightarrow \psi.$$

In general for our purposes we have the following **bridge theorem**:

Theorem (Maksimova 1979)

A logic L has:

1. *Deductive interpolation if and only if $\text{Alg}(L)$ has amalgamation;*
2. *Craig interpolation if and only if $\text{Alg}(L)$ has superamalgamation.*

CAVEAT: The two properties are often not equivalent. For intuitionistic logic, lax logic, and extensions of mHC, however, there is a **true deduction theorem**:

$$\phi \vdash_L \psi \iff \vdash_L \phi \rightarrow \psi.$$

Maksimova used this bridge theorem to show, for example:

Theorem

There are exactly seven consistent superintuitionistic logics L (i.e., $L \supseteq \text{IPC}$) which have interpolation.

To make algebras more concrete, and more easy to manipulate in general, we make use of [duality theory](#).

To make algebras more concrete, and more easy to manipulate in general, we make use of [duality theory](#).

Definition (Esakia space)

A partially ordered topological space $(X, \leq, \tau)$ is called an **Esakia space** if:

1. (X, τ) is compact;
2. (PSA) Whenever $x \not\leq y$ there is a clopen upwards-closed set U such that $x \in U$ and $y \notin U$;
3. Whenever C is a clopen subset, $\downarrow C = \{x \in X : \exists y \in C, x \leq y\}$ is clopen.

To make algebras more concrete, and more easy to manipulate in general, we make use of **duality theory**.

Definition (Esakia space)

A partially ordered topological space $(X, \leq, \tau)$ is called an **Esakia space** if:

1. (X, τ) is compact;
2. (PSA) Whenever $x \not\leq y$ there is a clopen upwards-closed set U such that $x \in U$ and $y \notin U$;
3. Whenever C is a clopen subset, $\downarrow C = \{x \in X : \exists y \in C, x \leq y\}$ is clopen.

All **finite posets** can be seen as Esakia spaces. The natural morphisms between them are **continuous p-morphisms**:

Definition (p-morphism)

A continuous map $f: X \rightarrow Y$ between Esakia spaces is called a *p-morphism* if it is order-preserving and whenever $f(x) \leq y$ there is some $y' \geq x$ such that $f(y') = y$.

We have the categories **HA** of Heyting algebras and homomorphisms and **Esa** of Esakia spaces and continuous p-morphisms. Duality is the statement:

Theorem (Esakia 1974)

*The categories **HA** and **Esa** are dually equivalent, in symbols, $\mathbf{HA} \equiv^{\text{op}} \mathbf{Esa}$.*

¹Studied first, to the best of our knowledge, by (Palmigiano 2004).

We have the categories **HA** of Heyting algebras and homomorphisms and **Esa** of Esakia spaces and continuous p-morphisms. Duality is the statement:

Theorem (Esakia 1974)

*The categories **HA** and **Esa** are dually equivalent, in symbols, $\mathbf{HA} \equiv^{\text{op}} \mathbf{Esa}$.*

To take care of operators, we look at classes of [Esakia spaces with additional relations](#).

Definition (Modal Esakia space)

A tuple $(X, \leq, R, \tau)$ is a **modal Esakia space**¹ if:

1. $R = \leq \circ R \circ \leq$;
2. $R[x]$ is a closed upset for each $x \in X$;
3. Whenever U is a clopen upset, $R^{-1}[U]$ is a clopen upset.

Morphisms are **p-morphisms for both $\leq$ and R** .

We likewise have:

$$\mathbf{MHA} \equiv^{\text{op}} \mathbf{MEsa}.$$

¹Studied first, to the best of our knowledge, by (Palmigiano 2004).

The axioms of lax logic correspond to the following:

Proposition (Bezhanishvili and Ghilardi 2007)

A modal Esakia space $(X, \leq, R)$ is dual to a nuclear Heyting algebra if and only if:

$$xRy \iff \exists z(x \leq z \leq y \text{ and } zRz).$$

We call them **nuclear Esakia spaces**, and write:

$$S_x := \{z \in X : zRz\}.$$

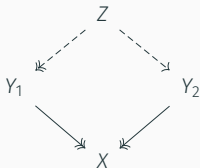
We systematically employ **duality** to connect amalgamation and a geometric property.

We systematically employ **duality** to connect amalgamation and a geometric property.

modal Heyting algebras $\equiv$ modal Esakia spaces
modal Heyting homomorphisms $\equiv$ modal p-morphisms
injective map $\equiv$ surjective map

We systematically employ **duality** to connect amalgamation and a geometric property.

modal Heyting algebras $\equiv$ modal Esakia spaces
modal Heyting homomorphisms $\equiv$ modal p-morphisms
injective map $\equiv$ surjective map



Co-amalgamation.

As an illustration we prove the following (originally proved, proof-theoretically, by Iemhoff):

Theorem (Iemhoff 2024)

Propositional Lax logic has the deductive interpolation property.

As an illustration we prove the following (originally proved, proof-theoretically, by Iemhoff):

Theorem (Iemhoff 2024)

Propositional Lax logic has the deductive interpolation property.

Proof.

Take $X \xrightarrow{f} Z \xleftarrow{g} Y$. Key observations:

1. We form the pullback:

$$X \times_Z Y := \{(x, y) : f(x) = g(y)\},$$

and equip it with the product order and product R relation.

Co-amalgamation for nuclear Esakia spaces

As an illustration we prove the following (originally proved, proof-theoretically, by Iemhoff):

Theorem (Iemhoff 2024)

Propositional Lax logic has the deductive interpolation property.

Proof.

Take $X \xrightarrow{f} Z \xleftarrow{g} Y$. Key observations:

1. We form the pullback:

$$X \times_Z Y := \{(x, y) : f(x) = g(y)\},$$

and equip it with the product order and product R relation.

2. By duality, and canonicity (see Adam's talk!) for amalgamation it suffices that the projections are still p -morphisms. They always are for the $\leq$ -relation.

Co-amalgamation for nuclear Esakia spaces

As an illustration we prove the following (originally proved, proof-theoretically, by Iemhoff):

Theorem (Iemhoff 2024)

Propositional Lax logic has the deductive interpolation property.

Proof.

Take $X \xrightarrow{f} Z \xleftarrow{g} Y$. Key observations:

1. We form the pullback:

$$X \times_Z Y := \{(x, y) : f(x) = g(y)\},$$

and equip it with the product order and product R relation.

2. By duality, and canonicity (see Adam's talk!) for amalgamation it suffices that the projections are still p -morphisms. They always are for the $\leq$ -relation.
3. Assume $x'Rx'$, and $x \leq x'$. It suffices to show that if $(x, y) \in X \times_Z Y$, there is some $y' \geq y$ such that $y'Ry'$. For this we use:
 - 3.1 The maps f and g are R -morphisms;
 - 3.2 We find $y_0 \geq y$ such that $g(y_0) = f(x')$;
 - 3.3 We use the fact that $R[y_0] \cap g^{-1}[f(x')]$ is closed and non-empty, and find a **maximal point there**. This point will be $y' \geq y$, and $y'Ry'$ by maximality.

Problem

Can we characterize – or show that there is no reasonable characterization – the extensions of PLL with interpolation?

Natural simplification: **Glivenko lax logic**:

$$\text{GPLL} := \text{PLL} \oplus j(jp \rightarrow p).$$

The proof outlined above is a generic blueprint used both in positive and negative results.

The proof outlined above is a generic blueprint used both in positive and **negative results**.

The construction $X \times_Z Y$ tends to provide the **generic shape of a counterexample**, as we will see now in the case of Fischer-Servi style logics.

Interpolation and Fischer-Servi logic

Definition

A modal Heyting algebra is a [Fischer-Servi algebra](#) if it also has a $\Diamond$ operator and satisfies:

1. $\Diamond(a \vee b) = \Diamond a \vee \Diamond b$ and $\Diamond 0 = 0$;
2. $\Diamond(a \rightarrow b) \wedge \Box a \leq \Diamond b$;
3. $\Diamond a \rightarrow \Box b \leq \Box(a \rightarrow b)$.

They form the algebraic semantics of the logic IK we saw before.

The dual frames and spaces are special: they carry a single relation, with compatibility conditions.

Definition (Fischer-Servi frames)

Let $F = (X, \leq, R)$ be a tuple where $(X, \leq)$ is a poset and R a binary relation. We say that F is an **FS-frame** provided the following conditions are satisfied:

- **F1:** $x' \geq xRy \implies \exists y'. x'Ry' \geq y$;
- **F2:** $xRy \leq y' \implies \exists x'. x \leq x'Ry'$.

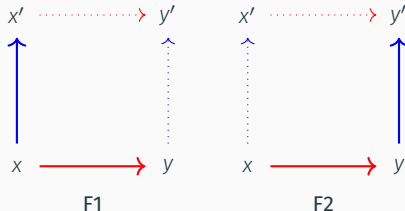


Figure 1: Confluence conditions for Fischer-Servi

Definition (FS-morphism)

Let $f: X \rightarrow Y$ be a map between **FS**-frames. We say that f is an **FS**-morphism if:

1. f is a $\leq$ -p-morphism.
2. If xRy , then $f(x)Rf(y)$.
3. If $f(x)Rz$, then $z \leq f(x')$ for some xRx' ;
4. If $f(x) \leq m$ and mRz , then there is some $x \leq x'$ and $x'R x''$ and $f(x'') \leq z$.

²There is a technical condition missing from what I presented, ask about it if you are dying to know, but it is not relevant for us today.

Definition (FS-morphism)

Let $f: X \rightarrow Y$ be a map between **FS**-frames. We say that f is an **FS**-morphism if:

1. f is a $\leq$ -p-morphism.
2. If xRy , then $f(x)Rf(y)$.
3. If $f(x)Rz$, then $z \leq f(x')$ for some xRx' ;
4. If $f(x) \leq m$ and mRz , then there is some $x \leq x'$ and $x'R x''$ and $f(x'') \leq z$.

Theorem

The categories $\mathbf{FS}_{\text{alg}}$ and $\mathbf{FS}_{\text{sp}}$ are dually equivalent².

²There is a technical condition missing from what I presented, ask about it if you are dying to know, but it is not relevant for us today.

Theorem (A., Bezhanishvili, Lemal, 2026)

The logics IK, IT, IK4, IS4 do not have interpolation.

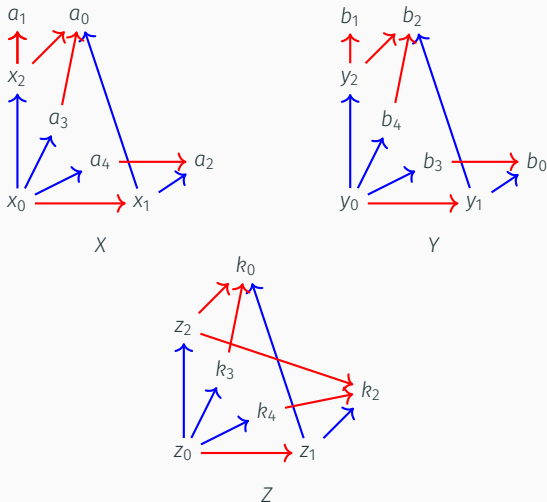
It suffices to provide an example of a diagram which cannot be amalgamated. So:

Refuting interpolation of Fischer-Servi

Theorem (A., Bezhanishvili, Lemal, 2026)

The logics IK, IT, IK4, IS4 do not have interpolation.

It suffices to provide an example of a diagram which cannot be amalgamated. So:



Refuting interpolation of Fischer-Servi

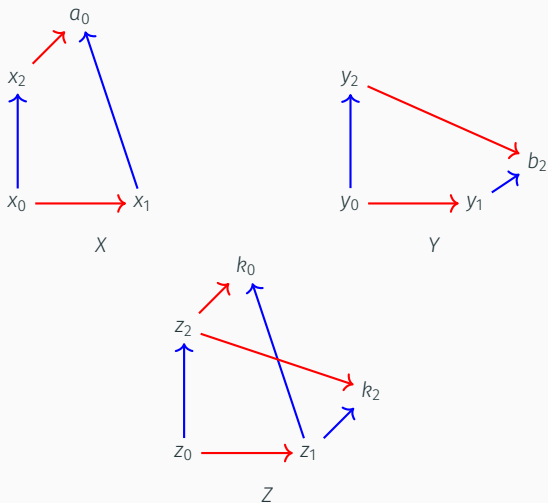


Figure 3: The co-V formation (Z, X, Y)

Refuting interpolation of Fischer-Servi

- We assume that W could amalgamate. Then we would have (w_0, w_1, w_2) mapping to x_0, x_1, x_2 .
- Since W satisfies F1, there is some point k . It ends up needing to be mapped to a_0 .
- By the diagram commuting, the triple maps also to y_0, y_1, y_2 , so k has to map to b_2 .
- But this is a contradiction.

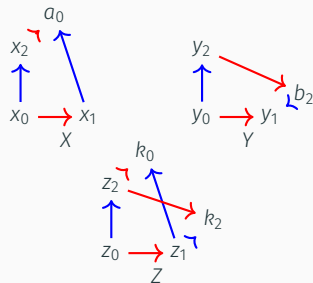


Figure 4: The co-V formation (Z, X, Y)

The same or minor modifications of this example show that also the following extensions of IK don't have interpolation:

$$IT := IK \oplus \Box p \rightarrow p \oplus p \rightarrow \Diamond p$$

$$IK4 := IK \oplus \Box p \rightarrow \Box \Box p \oplus \Diamond \Diamond p \rightarrow \Diamond p$$

$$IS4 := IK4 \oplus IT.$$

The same or minor modifications of this example show that also the following extensions of IK don't have interpolation:

$$\begin{aligned} \text{IT} &:= \text{IK} \oplus \Box p \rightarrow p \oplus p \rightarrow \Diamond p \\ \text{IK4} &:= \text{IK} \oplus \Box p \rightarrow \Box \Box p \oplus \Diamond \Diamond p \rightarrow \Diamond p \\ \text{IS4} &:= \text{IK4} \oplus \text{IT}. \end{aligned}$$

Can symmetry somehow help? That is, will IS5 fare any better?

The system IS5, also called MIPC (Ono 1977):

$$IS4 \oplus \Diamond p \rightarrow \Box \Diamond p \oplus \Diamond \Box p \rightarrow \Box p.$$

The system IS5, also called MIPC (Ono 1977):

$$\text{IS4} \oplus \exists p \rightarrow \forall \exists p \oplus \exists \forall p \rightarrow \forall p.$$

The system IS5, also called MIPC (Ono 1977):

$$\text{IS4} \oplus \exists p \rightarrow \forall \exists p \oplus \exists \forall p \rightarrow \forall p.$$

Similar to how S5 is the one-variable fragment of first-order logic, MIPC provides the one-variable fragment of intuitionistic first-order logic.

The system IS5, also called MIPC (Ono 1977):

$$\text{IS4} \oplus \exists p \rightarrow \forall \exists p \oplus \exists \forall p \rightarrow \forall p.$$

Similar to how S5 is the one-variable fragment of first-order logic, MIPC provides the one-variable fragment of intuitionistic first-order logic.

Theorem (G. Bezhanishvili and Marx, 2000(?))

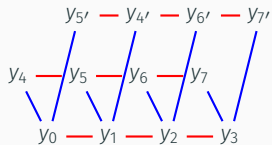
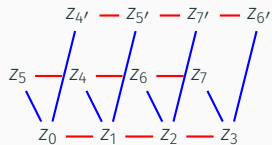
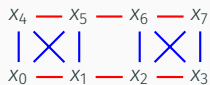
The system MIPC does not have interpolation.

Proof.

Via failure of amalgamation, in the next slide.



The counterexample:


 $\mathcal{G}$

 $\mathcal{H}$

 $\mathcal{F}$

To recover interpolation, one can look towards smaller logics – giving a larger number of models that can be amalgams – or towards extensions.

To recover interpolation, one can look towards smaller logics – giving a larger number of models that can be amalgams – or towards extensions.

Towards extensions, of course making the intuitionistic logic classical recovers the classical results. But beyond this, even for MIPC, the situation is very delicate.

To recover interpolation, one can look towards smaller logics – giving a larger number of models that can be amalgams – or towards extensions.

Towards extensions, of course making the intuitionistic logic classical recovers the classical results. But beyond this, even for MIPC, the situation is very delicate.

Let us consider Sol – **sollipsistic logic**:

$$\text{Sol} := \text{MIPC} \oplus \exists(p \wedge q) \leftrightarrow \exists p \wedge \exists q \oplus \forall(p \vee q) \leftrightarrow \forall p \vee \forall q.$$

Much like the strength axiom, these axioms are **classically trivial**.

To recover interpolation, one can look towards smaller logics – giving a larger number of models that can be amalgams – or towards extensions.

Towards extensions, of course making the intuitionistic logic classical recovers the classical results. But beyond this, even for MIPC, the situation is very delicate.

Let us consider Sol – **sollipsistic logic**:

$$\text{Sol} := \text{MIPC} \oplus \exists(p \wedge q) \leftrightarrow \exists p \wedge \exists q \oplus \forall(p \vee q) \leftrightarrow \forall p \vee \forall q.$$

Much like the strength axiom, these axioms are **classically trivial**.

Theorem

The logics Sol and $\text{Sol} \oplus \text{KC}$ have the Craig interpolation property.

To recover interpolation, one can look towards smaller logics – giving a larger number of models that can be amalgams – or towards extensions.

Towards extensions, of course making the intuitionistic logic classical recovers the classical results. But beyond this, even for MIPC, the situation is very delicate.

Let us consider Sol – **sollipsistic logic**:

$$\text{Sol} := \text{MIPC} \oplus \exists(p \wedge q) \leftrightarrow \exists p \wedge \exists q \oplus \forall(p \vee q) \leftrightarrow \forall p \vee \forall q.$$

Much like the strength axiom, these axioms are **classically trivial**.

Theorem

The logics Sol and $\text{Sol} \oplus \text{KC}$ have the Craig interpolation property.

Idea.

The axioms of sollipsistic logic forces the quantifiers into adjoints which **also exist dually** – an incredibly strong condition which allows the dual proofs to go through. $\square$

We are currently working towards extending this towards a **full characterization** of extensions of MIPC (or of appropriate subsystems).

Problem

Characterize – or show no reasonable such characterization exists – all extensions of MIPC with the interpolation property.

We are currently working towards extending this towards a **full characterization** of extensions of MIPC (or of appropriate subsystems).

Problem

Characterize – or show no reasonable such characterization exists – all extensions of MIPC with the interpolation property.

One can start with $\text{MGL} := \text{MIPC} \oplus \text{LC}$ – Monadic Gödel logic – where things should be a bit tamer.

Other properties beyond interpolation

Uniform definability of adjoints: the inclusion

$$\mathcal{F}_L(n) \hookrightarrow \mathcal{F}_L(n+1),$$

has left and right adjoints. In the work of (Pitts 1992) it was proved for Heyting algebras; for provability algebras by (Shavrukov 1993); recent proof-theoretic work has extended it to many other logics lemhoff 2019; lemhoff 2024; Giessen 2022; Giessen and Shillito 2026; Akbar Tabatabai, lemhoff, and Jalali 2021; Férée and Shillito 2026.

Uniform definability of adjoints: the inclusion

$$\mathcal{F}_L(n) \hookrightarrow \mathcal{F}_L(n+1),$$

has left and right adjoints. In the work of (Pitts 1992) it was proved for Heyting algebras; for provability algebras by (Shavrukov 1993); recent proof-theoretic work has extended it to many other logics lemhoff 2019; lemhoff 2024; Giessen 2022; Giessen and Shillito 2026; Akbar Tabatabai, lemhoff, and Jalali 2021; Férée and Shillito 2026.

An example of what we have been looking at:

Uniform interpolation holds for all extensions of KM.3:

$$\begin{aligned}\text{KM} &:= \text{IK} \oplus p \rightarrow \Box p \oplus \Box p \rightarrow (q \vee q \rightarrow p) \oplus (\Box p \rightarrow p) \rightarrow p \\ \text{KM.3} &:= \text{KM} \oplus (p \rightarrow q) \vee (q \rightarrow p).\end{aligned}$$

The logic KM.3 itself is not locally tabular, and the proof of uniform interpolation is one of the simplest we know.

Uniform definability of adjoints: the inclusion

$$\mathcal{F}_L(n) \hookrightarrow \mathcal{F}_L(n+1),$$

has left and right adjoints. In the work of (Pitts 1992) it was proved for Heyting algebras; for provability algebras by (Shavrukov 1993); recent proof-theoretic work has extended it to many other logics lemhoff 2019; lemhoff 2024; Giessen 2022; Giessen and Shillito 2026; Akbar Tabatabai, lemhoff, and Jalali 2021; Férée and Shillito 2026.

An example of what we have been looking at:

Uniform interpolation holds for all extensions of KM.3:

$$\begin{aligned}\text{KM} &:= \text{iK} \oplus p \rightarrow \Box p \oplus \Box p \rightarrow (q \vee q \rightarrow p) \oplus (\Box p \rightarrow p) \rightarrow p \\ \text{KM.3} &:= \text{KM} \oplus (p \rightarrow q) \vee (q \rightarrow p).\end{aligned}$$

The logic KM.3 itself is not locally tabular, and the proof of uniform interpolation is one of the simplest we know.

Is uniform interpolation in KM characterizable?

Ruitenburg property and fixed-point elimination: in (“On the Period of Sequences (An(p)) in Intuitionistic Propositional Calculus” 1984) it was shown that for **any formula** $\phi(p, \bar{q})$, there is an N such that:

$$\phi(p, \bar{q})^N \leftrightarrow \phi^{N+2}(p, \bar{q}).$$

This property is known to fail for almost all reasonable logics (Litak 2024); however, its presence says that endomorphisms and substitutions behave unexpectedly nicely – it has applications in **fixed-point intuitionistic modal logics** (Afshari and Grotenhuis 2025).

A quick mention of some other results

Ruitenburg property and fixed-point elimination: in (“On the Period of Sequences (An(p)) in Intuitionistic Propositional Calculus” 1984) it was shown that for **any formula** $\phi(p, \bar{q})$, there is an N such that:

$$\phi(p, \bar{q})^N \leftrightarrow \phi^{N+2}(p, \bar{q}).$$

This property is known to fail for almost all reasonable logics (Litak 2024); however, its presence says that endomorphisms and substitutions behave unexpectedly nicely – it has applications in **fixed-point intuitionistic modal logics** (Afshari and Grotenhuis 2025).

Litak (Litak 2024) asked if PLL has this property. We conjecture that yes; we have been able to show it for:

$$\text{GPLL} = \text{PLL} \oplus j(jp \rightarrow p),$$

by using a **translation**. But a deeper model-theoretic analysis of this property is also lacking.

Final thoughts

Some very vague and general questions, which I hope some of you might know how to answer:

Some very vague and general questions, which I hope some of you might know how to answer:

1. Interpolation is part of a web of interrelated properties – uniform interpolation, fixed-point elimination, Ruitenburg property, coequivalence separation, finitariness of unification – which concern definability, and appear to share a general toolbox. Is there a deeper way of seeing their interrelation?

Some very vague and general questions, which I hope some of you might know how to answer:

1. Interpolation is part of a web of interrelated properties – uniform interpolation, fixed-point elimination, Ruitenburg property, coequivalence separation, finitariness of unification – which concern definability, and appear to share a general toolbox. Is there a deeper way of seeing their interrelation?
2. Why do **classically trivial** axioms yield, in intuitionistic logic, **metalogically interesting** systems?

Some very vague and general questions, which I hope some of you might know how to answer:

1. Interpolation is part of a web of interrelated properties – uniform interpolation, fixed-point elimination, Ruitenburg property, coequivalence separation, finitariness of unification – which concern definability, and appear to share a general toolbox. Is there a deeper way of seeing their interrelation?
2. Why do **classically trivial** axioms yield, in intuitionistic logic, **metalogically interesting** systems?
3. Can we use the insights of these and other areas to *calibrate logical systems* – fine-tune them in terms of possessing or lacking a given property?

Thank you!
Questions?

References



Afshari, Bahareh and Lide Grotenhuis (2025). “Intuitionistic

μ

-Calculus with the Lewis Arrow”. In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Springer Nature Switzerland, pp. 374–392. ISBN: 9783032060853. DOI: [10.1007/978-3-032-06085-3_20](https://doi.org/10.1007/978-3-032-06085-3_20). URL: http://dx.doi.org/10.1007/978-3-032-06085-3_20.



Akbar Tabatabai, Amirhossein, Rosalie Iemhoff, and Raheleh Jalali (2021). “Uniform Lyndon Interpolation for Basic Non-normal Modal Logics”. In: *Logic, Language, Information, and Computation*. Springer International Publishing, pp. 287–301. ISBN: 9783030888534. DOI: [10.1007/978-3-030-88853-4_18](https://doi.org/10.1007/978-3-030-88853-4_18). URL: http://dx.doi.org/10.1007/978-3-030-88853-4_18.



Bezhanishvili, Guram and Silvio Ghilardi (2007). “An algebraic approach to subframe logics. Intuitionistic case”. In: *Annals of Pure and Applied Logic* 147:1-2, pp. 84–100. ISSN: 0168-0072. DOI: [10.1016/j.apal.2007.04.001](https://doi.org/10.1016/j.apal.2007.04.001). URL: <http://dx.doi.org/10.1016/j.apal.2007.04.001>.



Esakia, Leo (1974). In: 214.2, pp. 298–301.

-  Esakia, Leo (Jan. 2006). **“The modalized Heyting calculus: a conservative modal extension of the Intuitionistic Logic”**. In: *Journal of Applied Non-Classical Logics* 16.3-4, pp. 349–366. ISSN: 1958-5780. DOI: [10.3166/jancl.16.349-366](https://doi.org/10.3166/jancl.16.349-366). URL: <http://dx.doi.org/10.3166/jancl.16.349-366>.
-  Ewald, W. B. (1986). **“Intuitionistic Tense and Modal Logic”**. In: *The Journal of Symbolic Logic* 51.1, pp. 166–179. ISSN: 00224812. URL: <http://www.jstor.org/stable/2273953> (visited on 08/18/2026).
-  Fairtlough, Matt and Michael Mendler (1997). **“Propositional Lax Logic”**. In: *Information and Computation* 137.1, pp. 1–33. ISSN: 0890-5401. DOI: <https://doi.org/10.1006/inco.1997.2627>. URL: <https://www.sciencedirect.com/science/article/pii/S0890540197926274>.
-  Férée, Hugo and Ian Shillito (2026). ***Pitts and Intuitionistic Multi-Succedent: Uniform Interpolation for KM***. DOI: [10.48550/ARXIV.2602.16445](https://arxiv.org/abs/2602.16445). URL: <https://arxiv.org/abs/2602.16445>.
-  Fischer-Servi, Gisèle (1984). **“Axiomatizations for some intuitionistic modal logics”**. In: *Rendiconti del Seminario Matematico - Polito* 42.3, pp. 179–194.

-  Giessen, Iris van der (2022). “Uniform Interpolation and Admissible Rules: Proof-theoretic investigations into (intuitionistic) modal logics”. PhD thesis. Utrecht University Library. URL: <http://dx.doi.org/10.33540/1486>.
-  Giessen, Iris van der and Ian Shillito (2026). *Uniform interpolation with constructive diamond*. URL: <https://arxiv.org/abs/2602.16880>.
-  Goldblatt, Robert I. (Jan. 1981). “Grothendieck Topology as Geometric Modality”. In: *Mathematical Logic Quarterly* 27.31-35, pp. 495–529. ISSN: 1521-3870. DOI: 10.1002/malq.19810273104. URL: <http://dx.doi.org/10.1002/malq.19810273104>.
-  Iemhoff, Rosalie (Nov. 2019). “Uniform interpolation and the existence of sequent calculi”. In: *Annals of Pure and Applied Logic* 170.11, p. 102711. ISSN: 0168-0072. URL: <http://dx.doi.org/10.1016/j.apal.2019.05.008>.
-  — (2024). “Proof Theory for Lax Logic”. In: *Dick de Jongh on Intuitionistic and Provability Logics*. Springer International Publishing, pp. 203–229. URL: http://dx.doi.org/10.1007/978-3-031-47921-2_8.
-  Kuznetsov, A. V. (1985). “On the propositional calculus of intuitionistic provability”. English. In: *Sov. Math., Dokl.* 32, pp. 18–21.

-  Kuznetsov, A. V. and A. Yu. Muravitsky (Mar. 1986). **“On superintuitionistic logics as fragments of proof logic extensions”**. In: *Studia Logica* 45.1, pp. 77–99. ISSN: 1572-8730. DOI: [10.1007/bf01881551](https://doi.org/10.1007/bf01881551). URL: <http://dx.doi.org/10.1007/BF01881551>.
-  Litak, Tadeusz (2024). **“Ruitenburg’s Theorem mechanized and contextualized”**. In: *Proceedings Twelfth Workshop on Fixed Points in Computer Science, FICS@CSL 2024, Naples, Italy, 19-20th February 2024*. Ed. by Alexis Saurin. Vol. 435. EPTCS, pp. 41–57. DOI: [10.4204/EPTCS.435.4](https://doi.org/10.4204/EPTCS.435.4). URL: <https://doi.org/10.4204/EPTCS.435.4>.
-  Maksimova, Larisa L. (1979). **“Interpolation Properties of Superintuitionistic Logics”**. In: *Studia Logica: An International Journal for Symbolic Logic* 38.4, pp. 419–428.
-  **“On the Period of Sequences $An(p)$ in Intuitionistic Propositional Calculus”** (1984). In: *The Journal of Symbolic Logic* 49.3, pp. 892–899. ISSN: 00224812. URL: <http://www.jstor.org/stable/2274142>.
-  Ono, Hiroakira (1977). **“On Some Intuitionistic Modal Logics”**. In: *Publications of The Research Institute for Mathematical Sciences* 13, pp. 687–722. URL: <https://api.semanticscholar.org/CorpusID:37783460>.
-  Palmigiano, Alessandra (2004). **“Dualities for some intuitionistic modal logics”**. In: *Liber Amicorum for Dick de Jongh*. Institute for Logic, Language and Computation, pp. 151–167.

-  Pitts, Andrew M. (Mar. 1992). “On an interpretation of second order quantification in first order intuitionistic propositional logic”. In: *Journal of Symbolic Logic* 57.1, pp. 33–52. URL: <https://doi.org/10.2307/2275175>.
-  Shavrukov, Vladimir Yu. (1993). *Subalgebras of diagonalizable algebras of theories containing arithmetic*. eng. Warszawa: Instytut Matematyczny Polskiej Akademii Nauk. URL: <http://eudml.org/doc/219353>.
-  Simpson, Alex K. (1994). “The Proof Theory and Semantics of Intuitionistic Modal Logic”. PhD thesis. University of Edinburgh.
-  Visser, Albert (1981). “A Propositional Logic with Explicit Fixed Points”. In: *Studia Logica: An International Journal for Symbolic Logic* 40.2, pp. 155–175. ISSN: 00393215, 15728730. (Visited on 08/18/2026).